3. **Nutation:** The growth of the stem apices occurs in a zig-zag manner. It is because the two sides of the stem apex alternatively grow more. Such growth movements are called as nutational movements. In some plants nutational movements allow the shoots apex to grow in helical path in upward direction. This movement is called **circumnutation**. It is commonly found in the stems of climbers of Cucurbitaceae (Figure 15.13).



Figure 15.13: Circumnutation

b. **Autonomic Movement of variation:** It happens in Indian telegraph plant. (*Desmodium gyrans*). The compound leaf consists of a larger terminal and two smaller lateral leaflets. During day time, the two lateral leaflets move upward at an angle of 90° and come to lie parallel to the rachis. Again, they may move downward at 180° so that they are parallel to the rachis. They may again move upward at 90° to come in their original position. All these movements occur with jerks after intervals, each movement being completed in about 2 minutes (Figure 15.14).



Figure 15.14 Autonomic Movement of Variation

ii. Paratonic (induces) movements of curvature

The movement arising from external stimulus is called **Paratonic** (Induced) movements of curvature. They are of two types. 1) Tropic movements 2) Nastic movements (Table 2)

a. Tropic movements

A movement that occurs in response to an unidirectional stimulus is called **tropic movement** or tropism. There are seven types in tropic movements (Geotropic, Phototropic, Thigmotropic, Chemotropic, Hydrotropic, Thermotropic and Aerotropic)

1. Geotropism

The movements which take place in response to gravity stimulus are called **geotropic movements**. The primary roots growing down into soil are **positive geotropic.** Primary stems that grow away from soil (against gravity) are **negative geotropic**. Secondary roots growing at right angles to the force of gravity are **Diageotropic**. Secondary lateral roots which grow obliquely downwards are **Plagiogeotropic**. Lateral roots and branches which are not sensitive to gravitational stimulus are **Apogeotropic**.

2. Phototropism

The tropic movement taking place as a response to light stimulus is called **phototropism**. Some of the plant parts such as stems, branches, leaves and pedicels of flowers move towards the stimulus of light and are said to be **positively phototropic**

Table 2: Differences between Tropic Movements and Nastic movements

Tropic Movements	Nastic movements
1. Movement occurs due to unidirectional stimulus.	These movements occur due to a diffused stimulus.
2. The stimulus acts on protoplasm from one direction only.	The stimulus acts on the protoplasm from all sides.
3. The response is directly related to the direction of the stimulus.	The response has no relation to the direction of the stimulus but with organ.
4. These are movements of curvature caused by unilateral growth.	These are also the movement of curvature but they are caused by reversible turgor changes.
5. Tropic movements may be phototropic, geotropic, hydrotropic, thigmotropic, chemotropic, thermotropic or aerotropic.	Nastic movements may be seismonastic, photonastic or thermonastic

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